

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12407773
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Fang; Feiyu et al.

DSL fault location

Abstract

A method of determining the location of a disconnection on a digital subscriber line, in particular a VDSL line, where the line has a number of nodes or connection points along it. A statistical model is generated from a population of lines that maps the loop (line) lengths of each of those lines obtained by line test measurements (such as single ended line test traces—SELT traces) against the corresponding loop lengths obtained from inventory data. The model is then used to determine a predicted loop length by mapping a measured loop length (taken from line test measurements) onto an inventory loop length using the model. Knowledge of the node positions on the line is then used to give a probability a line disconnect occurring at a given node by mapping the predicted loop length onto the node positions.

Inventors:	Fang; Feiyu (London, GB), Kekempanos; Lykourgos (London, GB), Gilks; Daniel (London, GB)
Applicant:	BRITISH TELECOMMUNICATIONS PUBLIC LIMITED COMPANY (London, GB)
Family ID:	75472555
Assignee:	BRITISH TELECOMMUNICATIONS public limited company (London, GB)
Appl. No.:	18/549298
Filed (or PCT Filed):	March 07, 2022
PCT No.:	PCT/EP2022/055699
PCT Pub. No.:	WO2022/189332
PCT Pub. Date:	September 15, 2022

Prior Publication Data

Foreign Application Priority Data

GB2103204Mar. 08, 2021

Publication Classification

Int. Cl.:H04M3/08 (20060101); H04M3/30 (20060101)

U.S. Cl.:

CPH04M3/085 (20130101); H04M3/306 (20130101);

Field of Classification Search

CPC:H04M (3/085); H04M (3/306)

USPC:379/22

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6950497	12/2004	Drury et al.	N/A	N/A
7359330	12/2007	Lunt et al.	N/A	N/A
8705636	12/2013	Fertner et al.	N/A	N/A
9071678	12/2014	Yang et al.	N/A	N/A
9130655	12/2014	Cunningham et al.	N/A	N/A
9148504	12/2014	Berg et al.	N/A	N/A
9246614	12/2015	Berg et al.	N/A	N/A
10051117	12/2017	Zahedi et al.	N/A	N/A
10523810	12/2018	Mohseni	N/A	H04B 3/487
11089150	12/2020	Dierickx	N/A	H04M 3/2209
2002/0072386	12/2001	Ginesi	455/507	H04L 5/023
2003/0048878	12/2002	Drury et al.	N/A	N/A
2003/0156685	12/2002	Erving	379/27.01	H04M 3/30
2006/0098725	12/2005	Rhee	375/222	H04L 12/2874
2006/0120442	12/2005	Melsa	375/222	H04B 3/493
2015/0030059	12/2014	Kalavai	375/224	H04M 3/306
2016/0087864	12/2015	Kerpez	370/242	H04B 3/487
2016/0337512	12/2015	Kalavai	N/A	H04M 3/30
2017/0180549	12/2016	Zahedi	N/A	H04L 41/0677
2018/0027113	12/2017	Mohseni	379/22.04	H04L 41/0631
2020/0274789	12/2019	Lyon et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
0025316	12/1999	GB	N/A
2367971	12/2001	GB	N/A
03084198	12/2002	WO	N/A
2016155955	12/2015	WO	N/A

OTHER PUBLICATIONS

International Preliminary Report on Patentability for PCT/EP2022/055699 dated Sep. 21, 2023 (8 pages). cited by applicant

International Search Report and Written Opinion of the ISA for PCT/EP2022/055699 mailed Jun. 30, 2022, 13 pages. cited by applicant

Combined Search and Examination Report for GB Application No. 2103204.0 dated Aug. 12, 2021, 7 pages. cited by applicant

“G.996.2: Single-ended line testing for digital subscriber lines”, International Telecommunication Union, 2018, 60 pages. cited by applicant

“Advanced xDSL Line Testing and Fault Location for Service Providers”, 2015, ASSIA, Incorporated, 13 pages. cited by applicant

M. Bharathi and S. Ravishankar, “A combined correlation TDR and FDR procedure for Single Ended Loop topology estimation in DSL,” 2011 IEEE International Conference on Signal Processing, Communications and Computing (ICSPCC), Xi'an, China, 2011, pp. 1-6, doi: 10.1109/ICSPCC.2011.6061717. cited by applicant

Communication under Rule 71(3) EPC issued in EP22713575.3, dated Apr. 23, 2025, 8 pages. cited by applicant

Primary Examiner: Tieu; Binh Kien

Attorney, Agent or Firm: Nixon & Vanderhye P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is the U.S. national phase of International Application No. PCT/EP2022/055699 filed Mar. 7, 2022 which designated the U.S. and claims priority to GB 2103204.0 filed Mar. 8, 2021, the entire contents of each of which are hereby incorporated by reference.

FIELD OF THE INVENTION

(2) This invention relates to a method of identifying the location of a fault on a digital subscriber line, in particular the location of a disconnection on a digital subscriber line.

BACKGROUND TO THE INVENTION

(3) Digital subscriber line (DSL) technology, often referred to as “broadband”, is a family of services that provides high speed digital data transmission over the metallic twisted copper pairs that form part of a local telephone network. DSL is commonly used to provide a customer's home with a network connection, typically to the Internet via an ISP.

(4) However, DSL lines are prone to faults, arising for example from faulty joints or interference. Various approaches have been used to identify and locate faults. For example, if a DSL line is synchronised and able to provide some level of service, then various DSL line parameters (e.g.

SNR margin, line attenuation, bit error rates) can be measured and analysed to identify certain of fault. These in-sync measurements are typically made at the DSLAM (digital subscriber line access multiplexer). Diagnostic exchange tests can also be done by specialist line test equipment located at the central office, and include copper line tests in the electrical domain. These include measurements such as capacitances and resistances, which can also be analysed to help identify faults.

(5) However, in very high speed DSL (VDSL) services, often referred to as “Fibre to the Cabinet”, specialist line test equipment is generally not available, as the DSLAM for VDSL services is typically not located in an exchange, but instead in a primary connection point (PCP) such as a roadside cabinet, which may not have the physical capacity for such equipment, or such equipment may be prohibitively costly to install in the cabinet.

(6) Furthermore, many of the measurements made by the DSLAM that are currently used to identify faults rely on a DSL line being initialised and synchronised. Therefore, certain faults such as a physically disconnected line make it impossible for a line to initialise and synchronise, thus making such measurements impossible.

(7) U.S. Pat. No. 7,359,330B2 describes a fault location system and method comprising two basic operations. The system provides an arrangement capable of finding the likely location of a fault in a telecommunications system effectively by comparing a test measurement on a line with a reference value and deriving from that comparison the likely location of the fault. Accordingly, the two operations are respectively obtaining one or more reference values to form the basis of the comparison, and carrying out the comparison itself.

(8) U.S. Pat. No. 695,049B2 describes a method of determining the location of a fault on a line in a telecommunications network which runs between first and second nodes, the line being connected to at least one further node along its length, comprising determining a plurality of possible routes for the line in dependence on the length of the line and the positions of the first and second nodes, selecting a route from the plurality of possible routes in dependence on the position of the at least one further node in relation to each of the plurality of possible routes, and determining the location of the fault along the selected route according to a length of line between the fault and one of said first and second nodes.

(9) International patent application WO2003084198A1 describes a method of detecting faults on a telephone line, the method comprising: comparing measured characteristics of the line with at least one model, said model(s) modelling expected characteristics of the telephone line; and in response to the comparison, generating a fault alert signal if the comparison between the measured characteristics and the modelled expected characteristics differ by more than a pre-determined threshold, wherein said characteristics comprise characteristics relating to the transmission of data on the line within a plurality of predetermined frequency bands.

SUMMARY OF THE INVENTION

(10) According to one aspect of the present invention, there is provided a method of determining the location of a disconnection on a test digital subscriber line, wherein said test digital subscriber line comprises a plurality of nodes, said method comprising: determining a measured loop length associated with each of the digital subscriber lines in a population of digital subscriber lines using line test measurements; determining an inventory loop length associated with each of the digital subscriber lines in the population of digital subscriber lines using inventory data; calculating a function that maps the measured loop lengths against the corresponding inventory loop lengths for each of the digital subscriber lines; determining a measured loop length associated with the test digital subscriber line using line test measurements; mapping the measured loop length associated with the test digital subscriber to a predicted loop length using the calculated function; determining a position for each of the plurality of nodes on the test digital subscriber line using inventory data; and determining the location of a disconnect by comparing the predicted loop length to the position of each of the plurality of nodes.

- (11) The method may further comprise: calculating a residual between the calculated function and the loop length measurements associated with each digital subscriber line in the population, and determining a distribution of the residuals; and wherein determining the location of a disconnect comprises calculating a probability of a disconnection at each node on the test digital subscriber line by mapping the predicted loop length onto the positions of each of the plurality of nodes using the distribution of the residuals.
- (12) Determining a measured loop length may comprise interpreting the line test measurements. The line test measurements may comprise single ended line test measurements, and interpreting the single ended line tests may comprise applying an inverse Fast Fourier Transform to the single ended line test measurements and determining a measured loop length using the results.
- (13) Calculating a function may comprise applying regression analysis to the measured loop lengths and inventory loop lengths. The function may be a linear function and the regression analysis linear regression analysis.
- (14) The nodes may comprise a joint or other connection on the line.
- (15) In a further aspect of the present invention, there is provided a test module for determining the location of a disconnection on a test digital subscriber line, wherein said test digital subscriber line comprises a plurality of nodes, said test module adapted in operation to: determine a measured loop length associated with each of the digital subscriber lines in a population of digital subscriber lines using line test measurements; determine an inventory loop length associated with each of the digital subscriber lines in the population of digital subscriber lines using inventory data; calculate a function that maps the measured loop lengths against the corresponding inventory loop lengths for each of the digital subscriber lines; determine a measured loop length associated with the test digital subscriber line using line test measurements; map the measured loop length associated with the test digital subscriber to a predicted loop length using the calculated function; determine a position for each of the plurality of nodes on the test digital subscriber line using inventory data; and determine the location of a disconnect by comparing the predicted loop length to the position of each of the plurality of nodes.
- (16) By combining the loop (line) length determinations with a strong prior belief that most network disconnects occur at cable joints/nodes, examples of the invention can utilise a series of lower confidence calculations to provide an improved prediction of an expected disconnection node when line sync cannot be achieved.
- (17) Specific examples use a linear regression model with normal distributions to determine probabilities of disconnect at the network nodes. Alternatively, distributions could be built as histograms between different sources of loop length, as well as optionally fitting a new normal distribution onto a generated histogram to calculate probabilities of a disconnect falling at each network node along a line.
- (18) The output is a most probable fault location from a number of nodes, and gives a list of probabilities of this fault falling at each node within the network, giving the further advantage of knowing the categories of engineer training required for a dispatched engineer and calculating which nodes in the network this engineer should investigate.
-

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a better understanding of the present invention reference will now be made by way of example only to the accompanying drawings, in which:
- (2) FIG. 1 is a system diagram showing a digital subscriber line running to a customer premises;
- (3) FIG. 2 is a high level flow chart summarising an example of the invention;
- (4) FIG. 3 is detailed flow chart summarising the steps of the model building stage in an example

of the invention; and

(5) FIG. 4 is detailed flow chart summarising the steps of the run-time stage in an example of the invention;

(6) FIG. 5 is a detailed flow chart summarising some alternative steps to calculate a probability of disconnect during run-time in an example of the invention;

(7) FIG. 6 is an example of a SELT UER trace;

(8) FIG. 7 is an example of a time-domain trace after applying an inverse FFT;

(9) FIG. 8 is an example of a graph plotting the inventory loop lengths versus the corresponding measured loop lengths derived from SELT UER;

(10) FIG. 9 is an example of a histogram of the residuals;

(11) FIG. 10 shows an example of Gaussian distributions generated with the vendor and SELT UER predictions of inventory loop length, plotted against node distances on a line;

(12) FIG. 11 shows a table of the resulting integrals/probabilities for nodes in an example of the invention.

DESCRIPTION OF PREFERRED EMBODIMENTS

(13) The present invention is described herein with reference to particular examples. The invention is not, however, limited to such examples.

(14) Examples of the present invention present a method of determining the location of a disconnection on a digital subscriber line, in particular a VDSL line, where the line has a number of nodes or connection points along it. A statistical model is generated from a population of lines that maps the loop (line) lengths of each of those lines obtained by line test measurements (such as single ended line test traces—SELT traces) against the corresponding loop lengths obtained from inventory data. The model is then used to determine a predicted loop length by mapping a measured loop length (taken from line test measurements) onto an inventory loop length using the model. Knowledge of the node positions on the line is then used to give a probability a line disconnect occurring at a given node by mapping the predicted loop length onto the node positions. The methods assume that a disconnected line effectively terminates the length of the line prematurely, and that disconnections are most likely to occur at existing joints or nodes.

(15) FIG. 1 is a simplified system diagram illustrating a telecommunications network **100** including a customer's premises **102**. The customer's premises **102** is connected to a DSLAM **104** located at a primary connection point PCP **106**, which is typically a roadside cabinet. The connection between the customer premises and DSLAM **104** is provided by a telephone line **108**, made of a pair of twisted copper or aluminium wires. Specifically, a network termination equipment NTE **110** is at the customer premises **102** end of the line **108**. The NTE **110** is often referred to as a line box or master socket, and is the demarcation point between the telephone network and the customer wiring in the customer premises **102**. The line **108** runs from the NTE **110** to a junction box **112**, and then onto a distribution point DP **114**. In this example, the DP **114** is a junction box located on a telephone pole **116**. The line **108** then continues onto the PCP **106** and specifically the DSLAM **104**. Within the customer premises **102**, the NTE **110** is connected to customer premises equipment CPE **118**, typically a router or home hub.

(16) A DSLAM is a network element that provides digital subscriber line (DSL) services to connected lines and associated customer premises. The line **108** is thus also referred to as digital subscriber line, or DSL line. The DSLAM **104** also has an onward connection, typically a fibre optic connection, to the local exchange **120**, and from there onto data provisioning networks **122** via suitable connections and equipment. The data provisioning networks may include the internet and other networks. A skilled person will appreciate that there are other elements in the network **100** that have been omitted for simplicity, such as elements that provide standard PSTN services to the line **108**.

(17) Also shown connected to the data provisioning network is a test module or test server **124**. The test module **124** comprises a processor and a data store, such as hard disk array or similar. The test

module **124** gathers data from the DSLAM **104**, and the processor can use that data, together with other data, to determine if there is a disconnect on the line.

(18) Examples of the invention try to identify the probability of a disconnect occurring at a particular node on the line. Nodes are typically cable joints or connection points along the line, where sections of cable are joined to form a complete line. Referring to FIG. **1**, the line **108** in this example has nodes at: DSLAM **104**, PCP junction **126**, footway box **128**, DP **114**, junction box **112**, and NTE **110**. On other lines there may be additional nodes, or conversely, some of these nodes might not be present.

(19) Whilst the present example shows a DSLAM residing in a PCP (roadside cabinet), the invention would still be applicable to configurations where the DSLAM is situated somewhere else. For example, the invention could still be applied to networks where the DSLAM is located in the local exchange **120**.

(20) Furthermore, FIG. **1** only shows a single line and associated elements. However, in practice there will be a number of lines, each serving a respective customer premises, connected to the DSLAM. Moreover, there will be many DSLAMs connected to the exchange, and nationally there will be many exchanges. Any number of these lines can be tested by the test server **124** using the methods described below.

(21) FIG. **2** is a flow chart summarising the general steps of the present invention as performed by the test module **124**.

(22) In step **200**, a statistical model is generated from a population of lines that maps the loop lengths of each of those lines obtained by line test measurements against the corresponding loop lengths obtained from inventory data (note, in the term loop length and line length are used interchangeably in this specification). The loop length obtained from line test measurements would typically be from single ended line test (SELT) measurements taken at the DSLAM. The inventory loop lengths may be determined from inventory records of cable lengths, which may be based on known cable lengths when cables were installed. The loop lengths are effectively measured with respect to the DSLAM.

(23) In step **202**, the loop length of a test line is measured (for example, using SELT measurements), and the model used to map the measured loop length onto an inventory loop length to give a predicted inventory loop length.

(24) In step **204**, the positions of the nodes on the test line are determined. These may be found from inventory records established when the line was installed. The node positions may be presented as distances from the DSLAM.

(25) In step **206**, a probability of a line disconnect occurring at each node is determined by mapping the predicted inventory loop length onto the node positions.

(26) Thus the method effectively employs a 2-stage process. There is a model building first stage (step **200**), and a run-time second stage (steps **202-206**).

(27) There now follows a more detailed example of the present invention as summarised by the flow chart of FIG. **3**, FIG. **4**, and FIG. **5**. FIG. **3** shows the steps of performed by the test module **124** during the model building stage, FIG. **4** shows the steps performed by the test module **124** in the run-time stage, with FIG. **5** showing an alternative for one of the run-time stage steps in FIG. **4**.

(28) Referring first to FIG. **3**, in step **300**, the test module **124** determines the measured loop length for each of a population of lines obtained from line test measurements. The population of lines can be one or more DSL lines operating in the network, and the data can be obtained over a period of time (for example over many days or weeks). In practice, a population including a large number of lines would be used.

(29) The loop lengths are obtained directly or indirectly from the DSLAMs serving the population of lines, which may include DSLAM **104**. Thus, the loop lengths are effectively distances measured from the DLSAM.

(30) Line test measurements are used to determine the loop lengths. One such line test

measurement is a SELT measurement, which can be triggered on a line by the DSLAM. A more detailed discussion of the SELT standard can be found in the International Telecommunication Union recommendation G.996.2, “Single-ended line testing for digital subscriber lines”. SELT measurements consist of sending wideband signals down a line and measuring the received signals reflected back from the line (over a frequency range).

(31) SELTs can be run remotely and can run regardless of the line synchronisation condition, so can be run on a line during pre-provisioning, and when the line is faulty or undergoing repairs. In contrast, service layer tests require a line to be in sync.

(32) The SELTs can return a line length directly as provided by the DSLAM vendor, and/or an SELT Uncalibrated Echo Response (UER) from a frequency sweep of the VDSL spectrum on the line.

(33) The SELT UER actually returns an array of complex values representing the phase and amplitude of the reflected waves detected over a wide range of frequencies. This is encoded as two arrays, Real and Imaginary components and a data scaling factor. An example of a SELT UER trace is shown in FIG. 6.

(34) The SELT UER is then fed through an inverse Fast Fourier Transform (FFT) function to turn the frequency-domain trace into a time-domain trace. This time-domain trace contains a large central peak (the outward signal detected as it leaves the DSLAM) and a smaller peak (the reflection signal) separated by a time difference. An example of the time-domain trace after applying the inverse FFT is shown in FIG. 7.

(35) The difference in these two peaks is multiplied by the signal speed to get the distance to the end of the line. This distance is actually the total distance from the DSLAM to the end of the line and back, so the distance is divided by two to give the loop length. In practice, we also run a smoothing filter on the FFT before detecting where the peaks are.

(36) The method employed by DSLAM vendors to generate the loop length is not made public, but is still based on running a SELT on the line and processing the results. Thus the vendor loop length and the SELT UER derived loop length both rely on processing line test measurements, specifically SELT measurements.

(37) Below shows a data set of a number of lines and the associated loop lengths (both vendor loop lengths and lengths obtained from a FFT of the SELT UER trace): Line 1: vendor 400 m, FFT 410 m Line 2: vendor 500 m, FFT 505 m Line 3: vendor 200 m, FFT 190 m Line 4 . . .

(38) Examples of the invention require at least one measured loop length, so either the vendor loop length, or the FFT loop length, or both could be used in this example. How these lengths are used will be described below.

(39) In step **302** the test module **124** determines the inventory loop lengths of the lines in the population of lines used in step **300**, and like in step **300**, the loop lengths are distances measured from the DSLAM. The inventory loop lengths may be determined from stored inventory data associated with the lines populated when the lines were installed. For example, the inventory data for a given line may contain explicit cable lengths, the location (coordinates) of each node, or the distances between each node. All this data can be used to determine the total loop length. For example, if the data is the distance between nodes, then the loop length can be approximated to the sum of these distances. The inventory data is thus effectively a long term value that has been stored in a database saying what the expected line length should be, in contrast to the measured loop lengths from step **300** representing a current/live length.

(40) Below is an example of inventory loop lengths for a number of lines: Line 1: 390 m (obtained from node to node distances 50+160+70+90+20) Line 2: 510 m (explicitly recorded in inventory data) Line 3: 190 m (obtained from node to node distances 60+25+30+40+35) Line 4 . . .

(41) In step **304**, the test module **124** builds a statistical model that maps the measured loop lengths (from step **300**) against the corresponding inventory loop lengths (from step **302**). The aim is to build a statistical model that can be used to determine an inventory line length when presented with

a measured line length from a line under test.

(42) As described in step **300**, the loop length obtained from line test measurements could be obtained in various ways e.g. vendor provided loop length or SELT UER derived loop length. The statistical model built at this stage is done using one of these determined sets of lengths, for example either using vendor loop length or SELT UER derived loop length. Thus, the resulting model would map the measured loop length, as provided by the vendor or as derived from SELT UER, onto the inventory loop length.

(43) The statistical model can be generated in a number of ways. One way is to use linear regression analysis to generate a linear model and fit a line $y=Ax$, with the inventory loop length set to the y values and the measured loop lengths set to the x values.

(44) FIG. **8** shows a graph plotting the inventory loop lengths versus the corresponding measured loop lengths derived from SELT UER for an example population of lines. The line plotted on graph has been fitted using linear regression analysis. Each data point in FIG. **8** represents a line from the population of lines. The residuals (the vertical differences between each data point and the regression line) follow a Gaussian distribution with a mean and standard deviation.

(45) FIG. **9** shows a histogram of the residuals. A Gaussian curve is shown fitted to the histogram illustrating the overall spread of residuals with a standard deviation around the mean (which is typically around zero).

(46) The linear regression analysis shown can be performed over the entire data set, or over a limited range of values for x .

(47) Whilst the example above has used the SELT UER loop lengths from step **300**, the vendor loop lengths could have been used instead or additionally to give a similar mapping using linear regression analysis.

(48) The output of the model at the most basic is thus the function $y=Ax$ and a standard deviation, where A and the standard deviation are determined using linear regression analysis of the measured and inventor loop lengths. This function can be used to predict an inventor loop length (y), given a measured loop length (x).

(49) However, the model can be further supplemented by population distribution analysis as follows.

(50) For the data within the model described above, let y =inventory loop length and $y'.sub.i=Ax.sub.i$, where $y'.sub.i$ is the SELT loop length prediction i . Population-based distribution analysis can then be performed over the set of values ($y, y'.sub.i$). We model this distribution as parameterised by a gradient of the best fit line and a standard deviation of the normal distribution.

(51) Optionally, this analysis can be performed between defined data ranges in $x.sub.i$, giving a gradient and standard deviation for each of the ranges, such as the 0th-20th, 20th-40th, 40th-60th, 60th-80th, and 80th-100th percentiles (in $x.sub.i$). Such an approach allows a non-linear relationship between the measured and inventory loop lengths to be modelled. For example, shorter loops might map slightly differently to longer loop lengths. Another way of tackling non-linear mappings is to use some other form of regression analysis that is non-linear.

(52) Below shows examples of each of two models that are generated—one built from SELT UER derived loop lengths measurements and one using vendor loop lengths: SELT UER linear model: gradient $A=1.02$, standard deviation=15 Vendor linear model: gradient $A=1.01$, standard deviation=20

(53) The resulting model or models are stored in step **306** for use during the run-time stage. The example models above can be stored as pairs of means and standard deviations for each SELT line length measuring method. Optionally, the each model may be further made up of separate models for each defined data range if analysis was performed over ranges of percentiles for $x.sub.i$.

(54) The models can be used to map a measured line loop length onto a predicted inventor loop length, which in turn is then used to map onto node positions to determine where a disconnect is likely to occur, and even determining a disconnect probability occurring at each node.

(55) FIG. 4 shows the steps performed by the test module **124** during the run-time stage, which can be run at any time, either reactively (e.g. in response to a customer reporting a fault), or proactively (e.g. without knowledge of any specific problem). FIG. 5 shows some alternative steps to some of those in FIG. 4 in another example of the invention.

(56) In step **400**, the test module **124** determines the measured loop length of a line **108** under test. As for step **300**, the loop length of the line **108** can be obtained directly or indirectly from the DSLAM **104** in the form of a vendor loop length or determined from a SELT Uncalibrated Echo Response (UER).

(57) In this example, the SELT UER length of the test line=330 m, and the vendor length=340 m.

(58) In step **402**, the statistical models generated in the model build stage and stored in step **306** are loaded. If more than one measured loop length is being used (for example, vendor loop length and SELT UER loop length models were generated in the model building stage, and both vendor loop length and SELT UER loop length can be determined for the test line), then all relevant associated models are retrieved.

(59) Following from the model build stage, the models loaded are as follow: SELT UER linear model: gradient $A=1.02$, standard deviation=15 Vendor linear model: gradient $A=1.01$, standard deviation=20

(60) In step **404**, the models retrieved from step **402** are used to map the measured loop lengths of the test line **108** from step **400** onto corresponding predicted inventory loop lengths. In this example, the measured loop length is 300 m, so: Predicted inventory loop length from SELT UER=330 m/1.02=324 m Predicted inventory loop length from vendor=340 m/1.01=337 m where 1.02 and 1.01 are the gradient A of the respective corresponding linear models.

(61) Mapping from a measured loop length onto a predicted inventory loop length confers several advantages. There could be incorrect scaling on the SELT lengths, which can result if the signal speed multiplier between the FFT time and distance is slightly off. Mapping also removes signal speed as a variable, as this scales the FFT length. There may also be hidden scalings in the vendor SELT lengths, so the mapping removes the need to trust calibrations on both SELT lengths. Furthermore, the current live length needs to be taken, but the node distances are taken from inventory measurements and are not measured “live”, so mapping the measured live length onto an inventory length before comparing with the inventory node distances yields more accurate results.

(62) In step **406**, the distance to each node (on line **108**) from the DSLAM **104** is determined. As described earlier with reference to FIG. 1, the nodes on line **108** are at the DSLAM **104**, PCP junction **126**, footway box **128**, DP **114**, junction box **112**, and NTE **110**. The distance to node can be determined using inventory data, which could include absolute distances between nodes, geographical coordinates of the nodes, etc. The result are a series of distances from the DSLAM **104** of each node, for example:

(63) TABLE-US-00001 Distance from Node DSLAM (m) DSLAM 0 PCP junction 5 Footway box 294 DP 319 Junction box 612 NTE 625

(64) At this stage, it is possible to estimate where a disconnect on the line could be located by comparing the predicted inventory loop length to the node distances. This is done in step **408**, by identifying the node whose distance from DSLAM is closest to the predicted inventory loop length. In this example, if the SELT UER measurements and model are used, the predicted inventory loop length is 324 m, which means that the closest node to that is the DP at 319 m. However, if the vendor measurements and model are used, then the predicted inventory loop length is 337 m, which means that the closest node to that is still the DP at 319 m. This approach gives a disconnect location as a single node.

(65) However, an alternative probabilistic approach can be used locate a disconnect resulting in a probability of a disconnect occurring at any given node. This approach is set out in FIG. 5, the steps of which provide an alternative to step **408**.

(66) In step **500**, which follows from step **406**, the model is used to generate a Gaussian probability

distribution function (PDF) that is combined with the node distances. This is done by positioning the mean (centre) of the PDF at the predicted inventory loop length, and further plotting that over the node distances. This is illustrated in FIG. 10, where the nodes positions are plotted as crosses, while the final NTE is plotted as a triangle. The node numbering and distance axis are centred on the DSLAM at zero as this is the location of the test equipment. The y-axis is the probability density, and the x-axis is the distance from DSLAM.

(67) 2 Gaussian PDFs are shown in FIG. 10—one associated with the SELT UER model 1002 and one associated with the vendor model 1004. A combined Gaussian PDF 1006 is also shown that combines both the SELT UER and vendor PDFs. Each of the individual PDFs or the combined PDF could be used to determine the location of a disconnect depending on what measurements and models were available.

(68) The combined Gaussian PDF may be generated as:

$$\mu = (\mu_{\text{sub.1}} + \mu_{\text{sub.2}}) / 2, \sigma_{\text{sup.2}} = (\sigma_{\text{sub.1}}^2 + \sigma_{\text{sub.2}}^2) / 4$$

where μ is the gradient of the combined PDF, $\mu_{\text{sub.1}}$ is the gradient of the first PDF, $\mu_{\text{sub.2}}$ is the gradient of the second PDF, σ is the standard deviation of the combined PDF, $\sigma_{\text{sub.1}}$ is the standard deviation of the first PDF, $\sigma_{\text{sub.2}}$ is the standard deviation of the second PDF.

(69) Using the example mean and standard deviations from the SELT UER and vendor models, a combined Gaussian PDF would result in $\mu = 330.5$, and $\sigma = 12.5$.

(70) In step 502, the percentage of the distribution per node is determined. This is done by integrating under the Gaussian PDF (SELT UER, vendor or combined) between the mid-points between the inventory node distances to give probabilities of disconnect at each node. The table in FIG. 11 shows an example of the resulting integrals (and thus probabilities) for each node.

(71) An alternative method is to integrate the distributions between fixed limits either side of a node location, such as node location ± 50 m.

(72) Then in step 504, the probability of disconnect per node is determined as the resulting integral at each node. For example, with reference to FIG. 11, the probability of disconnect using the SELT UER measurements are:

(73) TABLE-US-00002 DSLAM 0% PCP junction 0% Footway box 11.1% DP 89.9%
Junction box 0 NTE 0

(74) Thus, the most probably location of a disconnect is at the DP 114 followed by the footway box 128. A list in order of descending probability of disconnect can be generated, which can be used to prioritise the order of locations an engineer should be dispatched to for fault finding.

(75) As shown throughout the description, either one of the example loop length measurement approaches (SELT UER or vendor) and associated model can be used, or a combination of the two approaches and a combined model can be used.

(76) Exemplary embodiments of the invention are realised, at least in part, by executable computer program code which may be embodied in an application program data. When such computer program code is loaded into the memory of the processor 120 in the line analysis unit 118, it provides a computer program code structure which is capable of performing at least part of the methods in accordance with the above described exemplary embodiments of the invention.

(77) A person skilled in the art will appreciate that the computer program structure referred to can correspond to the flow chart shown in FIG. 2, where each step of the flow chart can correspond to at least one line of computer program code and that such, in combination with the processor 120 in the line analysis unit 118, provides apparatus for effecting the described process.

(78) In general, it is noted herein that while the above describes examples of the invention, there are several variations and modifications which may be made to the described examples without departing from the scope of the present invention as defined in the appended claims. One skilled in the art will recognise modifications to the described examples.

Claims

1. A method of determining the location of a disconnection on a test digital subscriber line, wherein said test digital subscriber line comprises a plurality of nodes, said method comprising: i) determining a measured loop length associated with each of the digital subscriber lines in a population of digital subscriber lines using line test measurements; ii) determining an inventory loop length associated with each of the digital subscriber lines in the population of digital subscriber lines using inventory data; iii) calculating a function that maps the measured loop lengths against the corresponding inventory loop lengths for each of the digital subscriber lines; iv) determining a measured loop length associated with the test digital subscriber line using line test measurements; v) mapping the measured loop length associated with the test digital subscriber to a predicted loop length using the calculated function; vi) determining a position for each of the plurality of nodes on the test digital subscriber line using inventory data; and vii) determining the location of a disconnect by comparing the predicted loop length to the position of each of the plurality of nodes.
 2. A method according to claim 1, further comprising: calculating a residual between the calculated function and the loop length measurements associated with each digital subscriber line in the population, and determining a distribution of the residuals; and wherein determining the location of a disconnect comprises calculating a probability of a disconnection at each node on the test digital subscriber line by mapping the predicted loop length onto the positions of each of the plurality of nodes using the distribution of the residuals.
 3. A method according to claim 1, wherein determining a measured loop length comprises interpreting the line test measurements.
 4. A method according to claim 3, wherein the line test measurements comprise single ended line test measurements, and interpreting the single ended line tests comprises applying an inverse Fast Fourier Transform to the single ended line test measurements and determining a measured loop length using the results.
 5. A method according to claim 1, wherein calculating a function comprises applying regression analysis to the measured loop lengths and inventory loop lengths.
 6. A method according to claim 5, wherein the function is a linear function and the regression analysis is linear regression analysis.
 7. A method according to claim 1, wherein each of said nodes comprises a joint.
 8. A test module for determining the location of a disconnection on a test digital subscriber line, wherein said test digital subscriber line comprises a plurality of nodes, said test module adapted in operation to: i) determine a measured loop length associated with each of the digital subscriber lines in a population of digital subscriber lines using line test measurements; ii) determine an inventory loop length associated with each of the digital subscriber lines in the population of digital subscriber lines using inventory data; iii) calculate a function that maps the measured loop lengths against the corresponding inventory loop lengths for each of the digital subscriber lines; iv) determine a measured loop length associated with the test digital subscriber line using line test measurements; v) map the measured loop length associated with the test digital subscriber to a predicted loop length using the calculated function; vi) determine a position for each of the plurality of nodes on the test digital subscriber line using inventory data; and vii) determine the location of a disconnect by comparing the predicted loop length to the position of each of the plurality of nodes.
-